

Lecture 5 - Transposes, Permutations & $\mathbb{R}^n$

→ In case we need to do row exchanges

$$A = LU$$

↓

$$\boxed{PA = LU}$$

$$\rightarrow \underline{P^{-1} = P^T}$$

$$P^T P = I$$

→ Transpose

$$A_{n \times m}^T = \boxed{A_{ij}^T = A_{ji}}$$

$$A^T \rightarrow m \times n$$

$$A \rightarrow n \times m$$

→ Symmetric matrices are matrices with equal transposes.

$$\boxed{A^T = A}$$

eg

$$\begin{bmatrix} 1 & 7 & 2 \\ -7 & 2 & 0 \\ 2 & 0 & 3 \end{bmatrix}$$

mirror image along diagonal

→ interesting point → AA^T will be a symmetric matrix.

eg $\begin{bmatrix} 1 & 3 \\ 2 & 3 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 4 \\ 3 & 3 & 1 \end{bmatrix}$

$$= \begin{bmatrix} 10 & 11 & 7 \\ 11 & 13 & 14 \\ 7 & 11 & 17 \end{bmatrix}$$

Proof → $(AA^T)^T$

$$\Rightarrow (A^T)^T \cdot A^T = AA^T$$

Hence $(AA^T)^T = AA^T$

Vector Spaces

→ $\mathbb{R}^2 \rightarrow$ xy plane for eg.

→ Axioms

- Any scalar multiplications should exist in same space.
- Additions must exist in the same space.

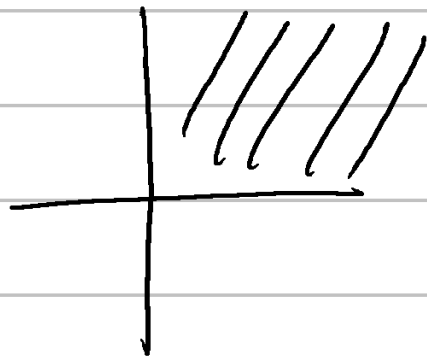
→ $\mathbb{R}^3$ vectors with all 3 real components.

$\begin{bmatrix} 3 \\ 2 \\ 0 \end{bmatrix}$ is a vector in $\mathbb{R}^3$

$\rightarrow \mathbb{R}^n \rightarrow \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}$ is a column vector in $\mathbb{R}^n$.

Counter example of vector space.

$\rightarrow \mathbb{R}^2$ with only $x > 0$ $y > 0$



$\rightarrow$ Addition works but scalar multiplications do not lie in the space. Non-closed

Subspace

→ A smaller subset of a vector space which satisfies all the rules of vector space is called a subspace.

eg vectors on a straight line in $\mathbb{R}^2$ that passes through origin.

All subspaces inside $\mathbb{R}^2$

① $\mathbb{R}^2$ itself.

② Lines passing through origin.

③ $\begin{bmatrix} 0 \\ 0 \end{bmatrix} \cong \boxed{\mathbb{Z}}$

→ Creating subspaces from matrices

$$A = \begin{bmatrix} 1 & 3 \\ 2 & 3 \\ 4 & 1 \end{bmatrix} \quad \text{cols in } \mathbb{R}^3$$

All their linear combinations form a subspace.

$$\rightarrow a \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} + b \begin{bmatrix} 3 \\ 3 \\ 1 \end{bmatrix}$$


called column space $C(A)$

basically the space created by the columns of a matrix

→ In this case the linear combinations of 2 vectors in $\mathbb{R}^3$ would give us a plane.